# Data Science - Python Project

Data Set download file - https://drive.google.com/file/d/1faPRfsnS8OMNu6c4BUjSY4cJV7axmPDs/view?usp=drive_link

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## 1. Background

SahajMobile is a EMI (easy monthly installment) phone company where users take out a loan to buy a phone and pay the phone back weekly or monthly. Each user has a payment schedule for their required payments.

## 2. Metric Definitions

One of the most important metrics that we use is MOIC (multiple on invested capital). Here are the formulas:

- **Payment MOIC** = Payment collected in the period ✚ Total advance of the cohort
- **Cumulative MOIC** = Cumulative payment collected to date ✚ Total advance of the cohort

Both metrics should be calculated separately. The chart should use **Cumulative MOIC**, while **Payment MOIC** may be included in the cohort summary table.

Example: Say a user gets a phone for 20,000 BDT and puts a downpayment of 6,000 BDT. Then they need to pay us back 14,000 BDT, and the total payment collected is 18,200 BDT. Therefore the MOIC is 18200/14000 = 1.3.

Example: if a cohort has a total advance of **100,000** and collects **10,000** in Month 1, then **Payment MOIC** for Month 1 is **0.10x** and **Cumulative MOIC** is also **0.10x**. If the same cohort collects another **15,000** in Month 2, then **Payment MOIC** for Month 2 is **0.15x**, but **Cumulative MOIC** becomes **0.25x**.

## 3. Task

Build a cohort-based MOIC curve from raw installment payment data using Python.
The purpose of this project is to analyze how payment collections build over time across different origination cohorts and to measure how quickly each cohort recovers its original advance.

## 4. Data Columns Available

- **Asset ID**
- **Origination Date**
- **Total Advance**
- **Total EMI**
- **Payment Date**
- **Payment Amount**

## 5. Required Logic

- Treat each **Asset ID** as one financed asset or loan
- Group assets by **origination month**
- Aggregate payments by **cohort** and **elapsed period**
- Calculate **cumulative collections** for each cohort
- Compute **MOIC** as cumulative collections divided by the original total advance of that cohort
- Plot **one vintage curve per origination month**

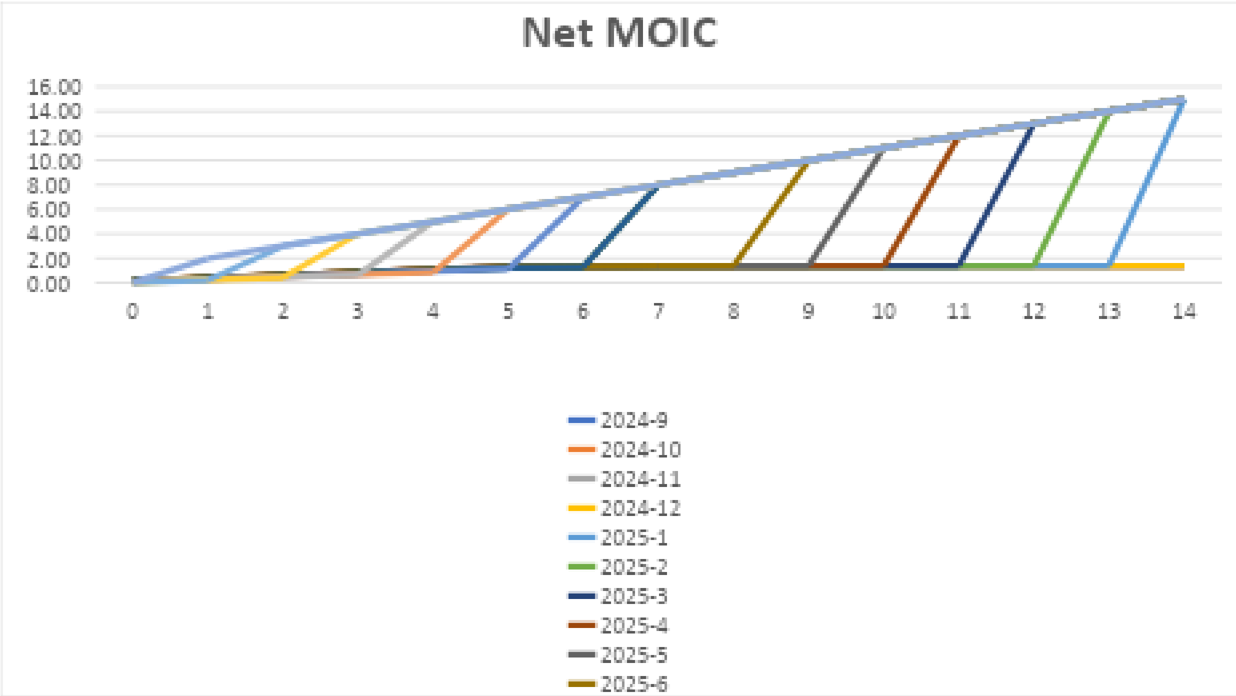
## 5. Chart Format

- **X-axis:** Months on Book or Installment Number
- **Y-axis:** Cumulative MOIC

## 6. Expected Output

- **Cleaned dataset**
- **Cohort summary table**
- **MOIC curve chart**

- **Python code for all of the above**




## 7. Notes

- Each **Asset ID** may appear in multiple rows because one loan can have many installment records.
- The file contains **duplicate records**, so exact duplicates should be removed before analysis.
- Any **invalid or abnormal payment dates** should be reviewed and excluded if necessary.
- The elapsed month should measure how far each payment is from the origination date.
- **Payment MOIC** shows the collection performance for one specific period only.
- **Cumulative MOIC** shows the total collection performance up to that point in time.